ML02 Values, names and types

Week 1, Python and a first prediction | Session 2 of 24 | Machine Learning

Today's question. What kind of thing is each part of a sale?

Objective. By the end: you can create whole numbers, decimals and text, and predict what happens when a name is given a new value.

Before this session: ML01 Your first instruction and your first prediction. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
value	One piece of information, such as 2000 or Juice.	One entry in a receipt.
integer	A whole number with no decimal part.	Counting 3 books.
float	A number with a decimal part.	A price of 2500.5 shillings.
string	Text inside quotation marks.	A name written on a label.
variable	A name that points to a value.	A label on a box of stock.
type	The kind of value: integer, float or string.	Knowing whether a box holds pens or juice.

Block A. Names point to values

A shop sells 3 items at UGX 2000 each. Run the cell and read each line of output.

Predict. Predict the three lines the cell will print. Write your guess on paper first.

Your guess, written before you run it

```
price = 2000
quantity = 3
cost = price * quantity
print(cost)
print(type(price))
print(type("2000"))
```

Run it, then compare. The cell prints 6000, then <class 'int'>, then <class 'str'>. price and quantity are variables. 2000 is an integer. The quoted 2000 is a string, which is text. The equals sign stores the value on its right under the name on its left.

Investigate.

1. Why is type("2000") a string?
2. What does the equals sign do in cost = price * quantity?

Modify. Change quantity to 5 and run the whole cell again.

Block B. A name keeps its value until you change it

Run the cell and watch what the printed total does when quantity changes.

Predict. Predict the three numbers printed. Will the second one change after quantity changes? Write your guess on paper first.

Your guess, written before you run it

```
price = 2500.5
quantity = 2
total = price * quantity
print(total)
quantity = 4
print(total)
total = price * quantity
print(total)
```

Run it, then compare. The cell prints 5001.0, then 5001.0, then 10002.0. 2500.5 is a float. Changing quantity does not change total until the line that calculates total runs again. Python works from the top of the cell to the bottom.

Investigate.

1. Why does the second print still show 5001.0?
2. What line makes the third number different?

Modify. Set quantity to 10, recalculate total and print it.

Make

Create a decimal price of 2500.5 and a quantity of 2. Print the total. Write one sentence on why "2500.5" with quotation marks is different.

Save your work as ML02. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
NameError: name 'price' is not defined	Python has not run a line that creates that name in this session.	Run the cell that creates the name first, and check the spelling and capital letters.
TypeError: can't multiply sequence by non-int of type 'str'	You tried to multiply two pieces of text.	Remove the quotation marks around the number.

Exit ticket

A variable is a name that points to a value, and the value stays until I store a new one.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.